

Sajed Albarah

JUNIOR SOFTWARE DEVELOPER

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PROFILE

First-class Computer Science and AI graduate with projects spanning interactive software, computer vision, natural language processing and formal verification. Built independent applications and worked in a five-person software engineering team. Strong analytical skills, practical experience with Python and JavaScript, and a willingness to learn unfamiliar tools through feedback and iteration.

EDUCATION

BSc Computer Science and AI, First Class | University of Sussex | 2023–2026

Selected module results: Databases 96% • Neural Networks 84% • Program Analysis 81% • Mathematics 91%

TECHNICAL SKILLS

Programming Python, JavaScript, Java, C and OCaml; introductory C# and R

Web and applications HTML, CSS, Bootstrap, Three.js, Unity; exposure to React and Microsoft Azure

Software practice Git/GitHub, debugging, testing, documentation, sprint planning and team reviews

Formal methods and tools Isabelle/HOL, algorithms, data structures, complexity analysis, VS Code, Codex, Vim and LaTeX

Data and machine learning SQL and MySQL, JSON, pandas, PyTorch, Jupyter, computer vision, NLP and model evaluation

SELECTED PROJECTS

PropertyTycoon | five-person software engineering project, 2025 | Unity, JSON

- Collaborated on a turn-based digital board game with property purchases, rent, cards, auctions, mortgages and bankruptcy. The team separated game state in JSON from systems implementing the rules.
- Contributed visual elements and the flow between the menu, player setup and game screens; refined asset placement and transitions to make the interface coherent during play.
- Worked through weekly sprint goals, team check-ins and shared documentation. Helped the group review integration issues and test the gameplay across different states.
- Co-wrote design and sprint documentation describing the UI, shared data and game-event systems, including the challenge of keeping the display aligned with changing game state.

Interactive Rubik's cube web app | personal project, 2026 | JavaScript, Three.js, Blender

- Built a browser-based 3D viewer for 3×3, mirror and Mastermorphix puzzles, with move input, animated turns, scramble, solve and reset controls.
- Connected cube move notation and a JavaScript solving library to the 3D animation state. Worked through rotation alignment, lighting and material issues across models exported from Blender.
- Created a model gallery and responsive pages, iterating on the interface and testing the interaction between user input, puzzle state and visual feedback.
- Separated viewer, move animation and page controls into distinct JavaScript modules so changes to one part of the interface were easier to reason about.

SELECTED PROJECTS CONTINUED

Sorting strategies formalisation | final-year research project, 2026 | Isabelle/HOL

- Modelled adjacent swaps and inversion-based sorting as relations over lists, using Isabelle/HOL to state precise properties of reductions and normal forms.
- Developed supporting definitions and lemmas about swap behaviour, preservation of list contents and inversion-count bounds; investigated how local commutation relates to sorting strategies.
- Studied the reverse-sorted list as a quadratic worst-case example, relating its inversion count to the minimum number of adjacent exchanges needed for sorting.
- Separated claims about the number of swaps from full running time, and wrote a technical report explaining the model, proof structure and limitations.
- Completed an 8,000 word technical report over one year.

CIFAR-10 image classification | machine learning project, 2026 | Python, PyTorch

- Designed and trained a convolutional neural network for ten-class image classification, with a fixed training/validation split and preprocessing based only on training data.
- Compared baseline training, data augmentation, weight decay and their combination across three random seeds; analysed accuracy, loss and train-validation gaps to assess overfitting.
- The augmentation condition achieved 78.77% mean validation accuracy and 78.07% mean test accuracy across the three runs.
- Kept the test set separate until model selection and computed normalisation statistics from the training partition, avoiding information leakage into evaluation.

Sentiment analysis on spam-rich text | Applied Machine Learning project | Python

- Prepared a noisy text dataset for sentiment classification, dealing with spam-heavy and inconsistent inputs before feature extraction and model training.
- Built and evaluated classification approaches on held-out data, then examined where spam and ambiguous language made predictions unreliable.
- Explained preprocessing, evaluation choices and limitations in the project report, rather than treating one accuracy score as the whole result.
- Used an n-gram logistic regression baseline to establish a clear comparison point for further text-classification experiments.

ADDITIONAL ACHIEVEMENT

- First place in a mathematics competition in Cairo, Egypt. 2022.
- Achieved top 0.1% percentile mark in standardized mathematics tests three times (ACT). 2022.
- Won a music scholarship at the University of Sussex twice for piano and organ performance. 2023 and 2024.